

Assignment 4 25F

1. Simplify the following Boolean function $F(A, B, C, D) = \sum(1, 2, 4, 7, 8, 9, 11)$, together with the don't-care conditions $d(0, 3, 5)$, and express the simplified Boolean function using:
 - a) NAND gates only, draw the logic diagram, and also provide algebraic method transform.
 - b) NOR gates only, draw the logic diagram, and also provide algebraic method transform.
2. A combinational circuit is defined by the equations
$$F1 = AB + A'B'C'$$
$$F2 = A + B + C'$$
$$F3 = A'B + AB'$$
Design a circuit which implements these three equations using a decoder and minimum number of NOR gates external to the decoder, and draw the logic diagram.
3. Design a sequence detector using JKFFs to recognize the occurrence of a particular sequence of bits (1101) (using Moore machine, non-overlapping mode), write down the necessary procedure for your design (No need to draw the logic diagram)
4. Design a sequence generator to generate the sequence 1011110 (starting with MSB) with minimum number of Flip-flops. Write down all the steps and draw the logic diagram
5. Design a counter with T flip-flops that goes through the following binary repeated sequence: 0, 1, 3, 7, 6, 4. Derive the input equations for the FFs. Show that when binary states 010 and 101 are considered as don't care conditions, the counter may not operate properly. Find a way to correct the design. (No need to draw the diagram)